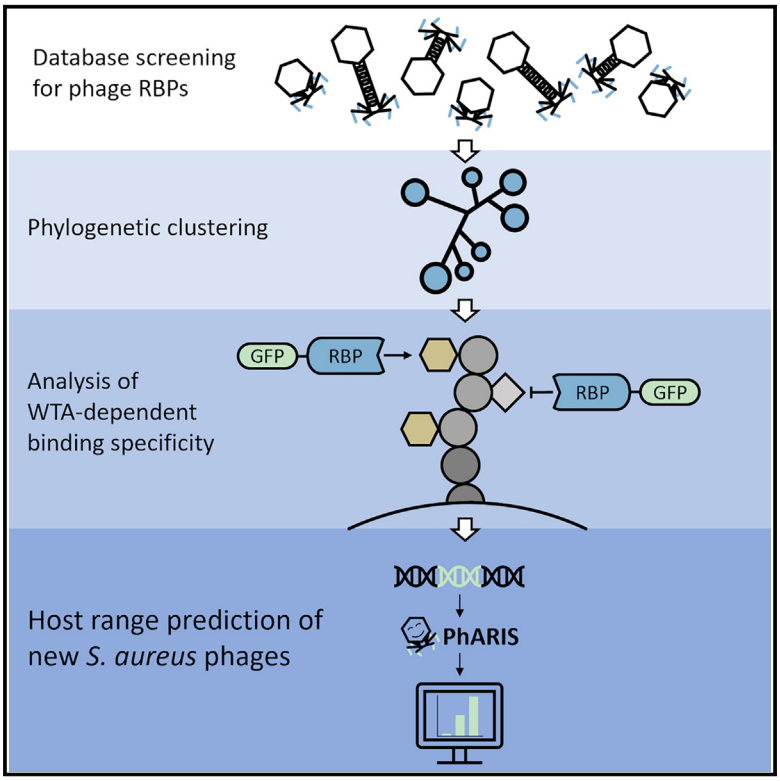
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Resource

Characterization and host range prediction of *Staphylococcus aureus* phages through receptor-binding protein analysis

Graphical abstract



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In brief

Bacteriophage infection requires successful binding by receptor-binding proteins (RBPs). Krusche et al. cluster *Staphylococcus aureus* phage RBPs and determine their cluster-specific binding patterns, allowing for prediction of phage binding specificities. This tool assists in the characterization of phages and their host range based on RBP sequences.

Highlights

- Receptor-binding proteins (RBPs) of 335 *S aureus* phages were identified
- Most *S. aureus* phages encode two distinct RBPs
- 15 recombinant RBPs bound to wall teichoic acid (WTA) polymers
- The new PhARIS tool predicts the host binding of new *S. aureus* phages

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Characterization and host range prediction of *Staphylococcus aureus* phages through receptor-binding protein analysis

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SUMMARY

Bacteriophages are crucial in bacterial communities and can be used for therapy of multidrug-resistant pathogens such as *Staphylococcus aureus*. However, the host range of new phages remains difficult to predict. We identified the receptor-binding proteins (RBPs) of 335 *S. aureus*-infecting phages, yielding 8 distinct RBP clusters. Recombinant representative RBPs of all clusters, including several subclusters, were analyzed for binding to *S. aureus* strains differing in potential phage receptor structures. Notably, most of the phages encoded two separate RBPs, and all RBPs used *S. aureus* wall teichoic acid (WTA) polymers as receptors, albeit with varying preference for WTA glycosylation patterns and backbone structures. Based on these findings, a sequence-based tool for predicting the adsorption of new phages was developed. Moreover, one of the RBPs proved useful for identifying *S. aureus*-type WTA in other bacterial species. These findings facilitate the characterization of phage and bacterial isolates and the development of phage therapies.

INTRODUCTION

Infections caused by antibiotic-resistant bacterial pathogens, including *Staphylococcus aureus*, threaten human health worldwide. Resistance to β-lactam antibiotics by methicillin-resistant *S. aureus* (MRSA) is difficult to overcome, as the remaining reserve antibiotics have much lower efficacy than β-lactams.[1] In 2019, *S. aureus* infections alone contributed to over 1 million deaths,[2] more than 700,000 of which were associated with resistance to antibiotics.[3] The urgent need for treatment strategies alternative to antibiotics has revived the interest in bacteriophages—viruses that infect and kill specific bacteria. Phage therapies have been used successfully against multiresistant bacterial pathogens[4] but are far from broad application, in part because of inherent difficulties in the prediction of bacterial susceptibility to specific phages. Most phages only have a limited host range, and we currently have no opportunities for predicting the specificity of a phage for different clonal lineages of *S. aureus*. The therapeutic potential of phages against infections by antibiotic-resistant bacteria also benefits from the narrow host range of phages, which decreases unwanted effects on potentially beneficial bacteria in a microbiome during treatment.[5]

All currently known *S. aureus* phages belong to the order Caudoviricetes and can be assigned to one of nine different genera. Rosenblumviruses (from the morphological group of podoviruses) are small phages with a very short, non-contractile tail and a mostly narrow host range. The morphological group of siphoviruses, with long, non-contractile tails, consists of the genera *Biseptimavirus* and *Peeveelvirus* (forming the subfamily Bronfenbrennervirinae), *Dubowvirus* and *Phietavirus* (subfamily Azeredovirinae), and *Triavirus*. The subfamily Twortvirinae, forming the morphological group of myoviruses with long, contractile tails includes the three *S. aureus*-infecting genera *Kayvirus*, *Silviavirus*, and *Baoshanvirus*.

Phage adsorption, the initial step of phage infection, depends on successful attachment of the phage to its host and, thus, on the specificity of the phage receptor-binding proteins (RBPs). Adsorption is often established in a two-step process.[6] The first contact between the phage and its host leads to reversible binding of one RBP to a receptor on the host surface. This initial step is followed by binding of a second RBP that irreversibly binds to the same or a second receptor,[7] which ultimately results in injection of viral DNA into the host. Only little is known about the molecular adsorption mechanism to *S. aureus*, and the RBPs of

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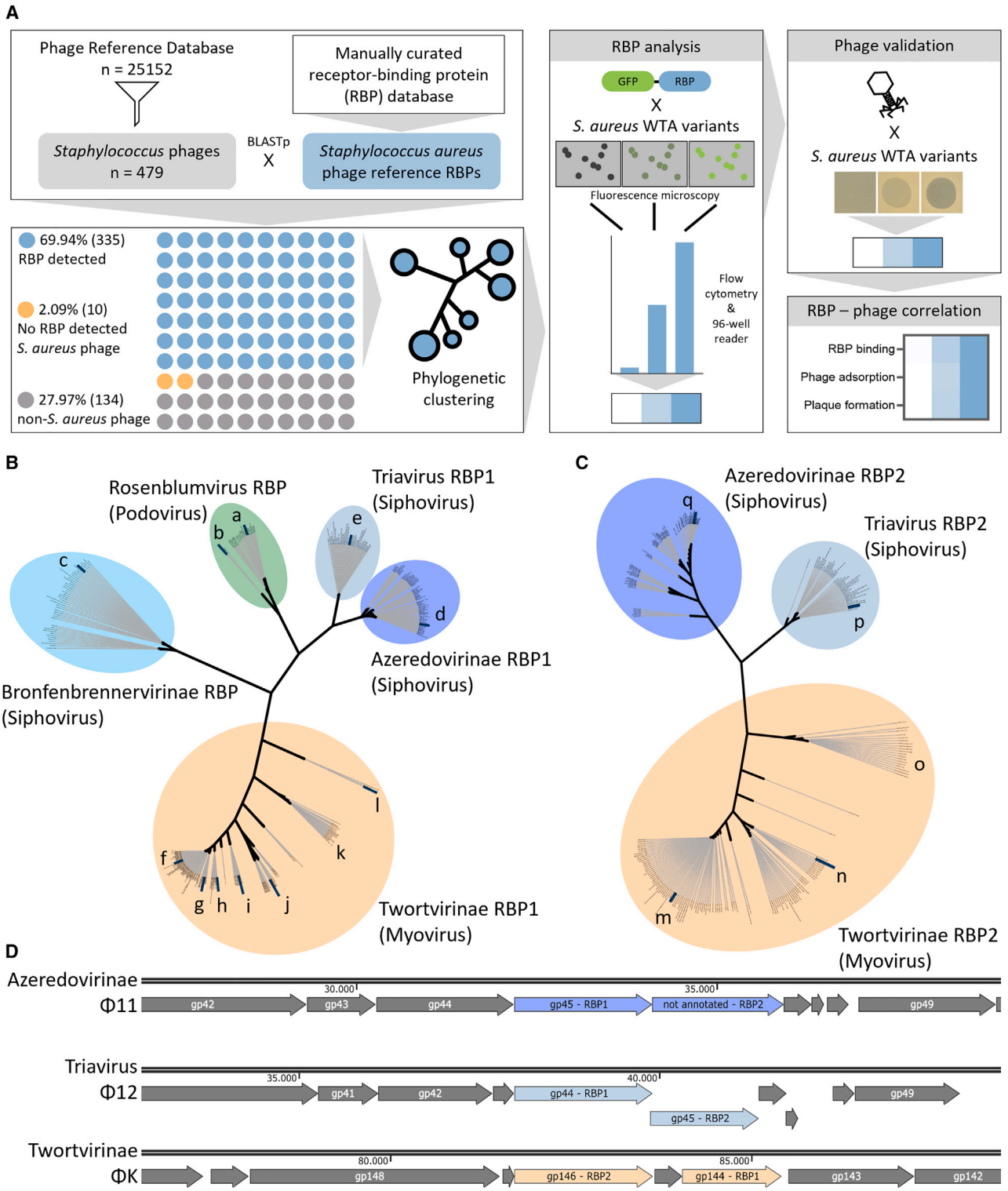


Figure 1. *S. aureus* phages encode one or two RBPs that show phylogenetic clustering with high intra-cluster homology

(A) Schematic of the bioinformatic (left) and wet lab (right) workflows that were used to classify and characterize *S. aureus* RBPs. A reference RBP panel, consisting of multiple typical representative RBPs, was aligned with a database of 479 *Staphylococcus* phage genomes from the INPHARED database, resulting in 335 phages with at least one RBP match (69.94%). In 10 *S. aureus* phage genomes (seven of those were giant viruses), no RBP could be detected (2.09%). The remaining 134 genomes belonged to non-*S. aureus* phages (27.97%), where no RBP could be detected. The resulting collection of *S. aureus* RBPs was clustered

(legend continued on next page)

S. aureus phages have not been identified, except for those of a few model phages. The RBP structures of the Azeredovirinae Φ11 and the closely related Φ80α have been determined by crystallization.[8,9] The Φ80α-RBP assembles into a large, multidomain homotrimer, and six RBP trimers are assembled around the baseplate core. Likewise, the full baseplate with integrated RBPs of *Rosenblumvirus* ΦP68 has been revealed.[10] For Twortvirinae, the binding behaviors of the two RBPs of ΦSA012 have been analyzed, although the exact RBP ligand specificities remain to be elucidated.[11] While phages of other bacterial taxa often use different receptors on host cells for initial and tight binding, such as lipopolysaccharides or membrane proteins,[12] *S. aureus* phages appear to use the same receptor for both steps: the peptidoglycan-linked wall teichoic acid (WTA) polymer consisting of ribitol phosphate (RboP) repeats.[13,14]

The RboP-WTA backbone of *S. aureus* is decorated with N-acetylglucosamine (GlcNAc) in various clone-specific conformations depending on the presence or absence of specific genome-encoded glycosyltransferases.[15] There are currently three RboP-WTA glycosyltransferases known in *S. aureus*. The main glycosyltransferase, TarS, mediates attachment of β-1,4-GlcNAc. It is present in the majority of *S. aureus* strains, while the accessory glycosyltransferases TarM (responsible for WTA α-1,4-GlcNAc modification) and TarP (WTA β-1,3-GlcNAc modification) can only be found in 37% and 7% of *S. aureus* strains, respectively.[16–20] Changes in WTA glycosylation have been shown to impact binding of and resistance to some phages.[21–23] *S. aureus* strains expressing TarM or TarP, for instance, are resistant to many rosenblumviruses.[16,21] In contrast to *S. aureus,* many non-*S. aureus* staphylococci (NASs) appear to carry WTA consisting of glycerol phosphate (GroP) repeat units with various sugar side chains, such as glucose or GlcNAc.[24,25] However, the WTA structures of most NAS species have remained unknown.

In this study, we aimed to elucidate the specificity determinants of *S. aureus*-phage interaction by identifying and analyzing the RBPs from most of the known *S. aureus* phage genera via combined computational and experimental approaches. Many of these phages appear to encode two different RBPs, which both bind to RboP-WTA, albeit with different affinities and variable preferences for WTA glycosylation types. Most RBPs of *S. aureus* phages can be grouped in several distinct clusters with high overall similarity that largely follow the phylogeny of the phages. The binding of recombinant RBPs matched that of the corresponding phages, substantiating the notion that the identified RBPs govern phage host range. By combining phylogenetic and binding specificity analysis of RBPs, we can now predict the binding pattern of nearly all known *S. aureus* phages. To this end, we developed an easy-to use tool called Phage Aureus RBP Identification System (PhARIS), enabling the user to predict RBPs from phage genomes and assess binding specificity of any *S. aureus* phage based on its genome sequence. Finally, we demonstrate that the Φ13 RBP can identify NAS strains carrying the virulence-associated RboP WTA.[26]

RESULTS

RBPs of *S. aureus* phages can be grouped into separate clusters according to amino acid sequence similarity

Phage infection requires adsorption to the host cell, but the phage determinants governing host-specific binding to *S. aureus* have remained only superficially understood. To elucidate the receptor specificities of *S. aureus* phages, all published information about RBP gene loci of *S. aureus* Azeredovirinae Φ80α and Φ11, Twortvirinae ΦSA012, and *Rosenblumvirus* ΦP68 and ΦS24-1, which encode partially characterized RBPs,[8–11,27] was collected. Additionally, new putative RBPs of *Triavirus* Φ12 and Bronfenbrennervirinae Φ13 and of multiple *S. aureus*-infecting Twortvirinae were discovered via HHpred, BLAST, and AlphaFold2 and by comparison of gene locations based on the fact that structural genes, including RBP genes, often show synteny in phage genomes.[28–30] The resulting curated list of RBPs was used to search for further RBP homologs in a database of 479 available *Staphylococcus* phage genomes, which was created by extracting all *Staphylococcus* phage genomes from the INPHARED phage database.[31] 335 of 479 phage genomes (69.94%) encoded proteins that matched at least one of the model RBPs with more than 65% identity and 65% overlap (Figure 1A). For 144 phages, no RBP1 gene could be assigned. Most of these were phages of NASs, suggesting strong differences between phage RBP1 proteins of *S. aureus* and NASs. According to the NCBI database, only 10 of the remaining 144 phages use *S. aureus* as their host, seven of which are giant viruses whose RBPs appear to be unrelated to those of other phage groups.

Following the identification of RBP1 genes, multiple alignment of protein sequences was performed with Clustal Omega,[32] visualizing the similarity of the different RBPs. The resulting phylogenetic tree (Figure 1B) revealed the existence of several distinct clusters of potential *S. aureus*-specific RBP1 proteins with very high intra-cluster similarity and very low similarity to the next-closest cluster. The cluster of RBP1 proteins from triaviruses was found to have the highest intrinsic relatedness, with a 97.8% identity of the query (Φ12-RBP1) to the most distant match of the cluster. The next closest match outside of this cluster was found in the Azeredovirinae RBP1 cluster, with an identity to the Φ12-RBP1 of 50.9%.

according to amino acid sequence homology. Per cluster, one or more green fluorescent protein (GFP)-RBP fusion proteins were designed, overexpressed, and analyzed for binding specificity. RBP binding results were compared to experiments of phage infection and adsorption.

(B) *S. aureus* phage RBP1s cluster according to their amino acid sequence. *Rosenblumvirus* RBPs could be separated into two subclusters (a, ΦP68-like; b, ΦCSA13-like), while siphoviruses could be separated into three different RBP clusters (c, Φ12-like [*Triavirus*]; d, Φ11-like [Azeredovirinae], and e, Φ13-like [Bronfenbrennervirinae]). Twortvirinae RBPs could be separated into seven different subclusters (f, ΦK-like; g, ΦStab20-like; h, ΦSA012-like; i, ΦPG-2021_10-like; j, ΦPG2021-17-like; k, ΦRemus-like; and k, ΦBS1-like).

(C) A second RBP (RBP2) was only detected in Twortvirinae (m, ΦK-like; n, ΦPG-2021_10-like; and o, ΦRemus-like), *Triavirus* (p, Φ12-like), and Azeredovirinae (q, Φ11-like). A second RBP was not found in rosenblumviruses and Bronfenbrennervirinae.

(D) Genomic location of RBP1 and RBP2 in one representative of each double-RBP-containing group (Azeredovirinae, *Triavirus*, and Twortvirinae).

Based on similarity, the *S. aureus* phage RBP1 proteins could be assigned to five main clusters with several subclusters. The *Rosenblumvirus* RBP1 proteins clustered into two distinct subclusters represented by the RBP1 proteins of ΦP68 (Figure 1B, subcluster a) and ΦCSA13 (Figure 1B, subcluster b), as described previously.[27] The Siphovirus RBP1 proteins clustered in three groups following their taxonomy (Figure 1B): those of Azeredovirinae (combining the genera of *Dubowvirus* and *Phietavirus*, represented by Φ11), *Triavirus* (represented by Φ12), and the more distant Bronfenbrennervirinae (combining *Biseptimavirus* and *Peeveelvirus*, represented by Φ13). The vast majority (99%) of the RBP1 proteins of currently known *S. aureus Rosenblumvirus*, *Dubowvirus*, *Phietavirus*, *Biseptimavirus*, *Peeveelvirus*, and *Triavirus* could be assigned to one of these four clusters.

S. aureus phages of the subfamily of Azeredovirinae consist of the two genera *Dubowvirus* and *Phietavirus*. While their RBP1 sequences branch into smaller subclusters among the Azeredovirinae RBP1 proteins, these do not reflect the phylogeny, with *Dubowvirus* and *Phietavirus* RBP1 being spread across the entire cluster. Bronfenbrennervirinae can also be separated into the two genera *Biseptimavirus* and *Peeveelvirus*. However, although the RBP1 proteins of these two genera are found in the same cluster, there is no clear subclustering between their sequences. Twortvirinae RBP1 proteins displayed higher heterogeneity compared to those of *Rosenblumvirus*, Azeredovirinae, Bronfenbrennervirinae, and *Triavirus*. For this reason, a 65% identity cutoff was used to filter out RBPs of NAS-infecting Twortvirinae, since this cutoff was found to effectively separate the RBP1 proteins of *S. aureus*-infecting Twortvirinae from those of NAS-infecting phages. RBP1 proteins with a BLAST-identity to the reference RBP1 (*Kayvirus* ΦK and *Silviavirus* ΦRemus) of 65% or higher were used for further analysis, while RBP1 proteins with BLAST-identity below 65% were discarded. A cutoff of 65% was found to include all RBP1 proteins from *S. aureus*-infecting Twortvirinae while simultaneously excluding those of most Twortvirinae that infect other staphylococci. The remaining *S. aureus*-specific Twortvirinae RBPs could be grouped into six subclusters. Three subclusters (Figure 1B, subclusters f–h) correspond to RBP1 of the genus *Kayvirus*, while phages with RBP1 of subclusters i and j did not have a specified genus. The remaining subclusters of Twortvirinae RBP1 proteins refer to *Silviavirus* (subcluster k) and *Baoshanvirus* (subcluster l).

In the Azeredovirinae Φ80α, a close relative of Φ11, the existence of a second RBP (RBP2) on its baseplate has been hypothesized previously, albeit without providing experimental evidence.[9] In the Twortvirinae ΦSA012, functional host adsorption has been demonstrated for two distinct RBPs, RBP1 and RBP2. While these two RBPs were found to have different binding properties during phage infection, the exact role of each RBP has remained unclear.[11] The RBP2 sequences were analyzed in a similar way as those of RBP1 proteins (Figure 1C). In the majority of rosenblumviruses as well as the Bronfenbrennervirinae, no RBP2 could be detected. In contrast, all triaviruses and Azeredovirinae carry a distinct, putative RBP2, which is encoded directly downstream of RBP1 (Figure 1D). Due to the organization of phage genomes into functional modules, genes encoding proteins of similar functions are often found in close proximity.[30] In fact, we found the putative RBP2 genes of Azeredovirinae (represented by Φ11), *Triavirus* (represented by Φ12), and Twortvirinae (represented by ΦK) to be located directly next to those of RBP1 (Figures 1C and 1D). Once again, the RBP2 proteins of the different genera of Azeredovirinae were found to be spread across the whole cluster, with no clear distinction between *Dubowvirus* and *Phietavirus* members. The Twortvirinae RBP2 sequences are separated into *Kayvirus* (subcluster m), undefined Twortvirinae (subcluster n), and *Silviavirus* (subcluster o).

We predicted homotrimeric structures of the C-terminal binding domains of all representative RBPs with the help of AlphaFold[33] (Figure S1). The 15 analyzed RBPs could be grouped according to six different tertiary structures. While the RBP1 trimers of *Rosenblumvirus*, *Triavirus*, and Azeredovirinae all had very similar predicted compact structures, reflecting those of the previously crystallized RBP1 of ΦP68[10] and Φ11[8], the structures of the Bronfenbrennervirinae and Twortvirinae RBP1 were predicted to be more elongated.

Overall, these results show a high intra-cluster conservation of RBP1 and RBP2 proteins across *S. aureus* phage taxa and partially even in between individual clusters. The conserved nature of the RBPs could be due to the highly conserved structure of *S. aureus* WTA, prompting us to investigate the binding of these proteins to the WTA of the host.

RBP1 of ΦK shows irreversible binding to WTA, while RBP2 binds reversibly to WTA, indicating the classic two-step binding pattern for *S. aureus* phages

To analyze the binding capability of the *in silico*-detected putative RBPs, the interaction of recombinant RBP proteins with bacterial strains expressing different WTA variants, the presumed RBP ligands, was investigated. To this end, a selection of RBPs fused to green fluorescent protein (GFP) (Table S2) were produced in *Escherichia coli*, purified, and tested for binding to *S. aureus* USA300 JE2, a methicillin-resistant strain that produces high levels of glycosylated RboP-WTA[34]; JE2 mutants lacking WTA (*tagO* mutant) or with altered WTA glycosylation (mutation of *tarS*, *tarM*, or both); as well as to *S. epidermidis* 1457 expressing GroP-WTA (Table 1). The fluorescence of the bacterium-adsorbed proteins was quantified via flow cytometry, and the dose-dependent binding of representative RBPs was confirmed (Figure S2). We focused initially on Twortvirinae ΦK because of its great potential as an agent for phage therapy.[35,36] RBP1 of ΦK was found to bind to *S. aureus*, irrespective of the glycosylation state of RboP-WTA, and to *S. epidermidis*, but it was not able to bind the WTA-deficient *S. aureus* Δ*tagO* mutant (Figures 2A and 2B).[37] Specifically, RBP1 of ΦK showed high affinity to *S. aureus* wild type (WT), Δ*tarM*, and Δ*tarS*, all carrying glycosylated RboP-WTA, as well as to the GroP-WTA expressing *S. epidermidis* and intermediary affinity to glycosylation-deficient *S. aureus* Δ*tarM* Δ*tarS*. Interestingly, phage K showed strong binding affinity to, and plaque formation on, *S. aureus* independent of the glycosylation type of the RboP-WTA. Weaker affinity was observed for S. *epidermidis* 1457 (Figure S3). To elucidate this discrepancy, the influence of the putative RBP2 on the host affinity of ΦK was assessed.

RBP2 of ΦK had a pronounced capacity to bind to all tested *S. aureus* strains except for the WTA-deficient Δ*tagO* mutant,

Table 1. *S. aureus* and *S. epidermidis* strains with different WTA backbone and glycosylation patterns used for RBP and phage binding experiments

Bacterial strain	WTA backbone and glycosylation
S. aureus USA300 JE2 WT	poly-RboP with GlcNAc largely in α-1,4 conformation to ribitol[23]
S. aureus USA300 JE2 Δ*tarM*	poly-RboP with GlcNAc linked in β-1,4 conformation to ribitol[18]
S. aureus USA300 JE2 Δ*tarS*	poly-RboP with GlcNAc linked in α-1,4 conformation to ribitol[17]
S. aureus USA300 JE2 Δ*tarM* Δ*tarS*	poly-RboP without glycosylation[17,18]
S. aureus USA300 JE2 Δ*tagO*	no WTA present[38,39]
S. epidermidis 1457	poly-GroP with glucose linked in α-1,2 conformation to glycerol[25]

which confirms its role as a WTA-binding RBP. The affinity of ΦK-RBP2 to the β-1,4-GlcNAc-glycosylated *S. aureus* Δ*tarM* and unglycosylated Δ*tarM* Δ*tarS* mutants was 2- to 3-fold higher compared to the parental USA300 JE2 WT strain and the Δ*tarS* mutant, which both predominantly carry α-1,4-GlcNAc (Figure S4). Interestingly, RBP2 of ΦK bound to *S. aureus* but dissociated quickly after washing of the RBP-bound bacterial cells, which was visible as a time-dependent decrease of fluorescence in the flow cytometer (Figures 2C and 2D). This behavior was not observed for RBP1 of ΦK. Thus, while RBP1 binds irreversibly, RBP2 appears to bind to WTA only in a reversible manner.

As we suspected that reversibly bound RBPs could get lost after washing of the bacteria during flow cytometry analysis, we developed an additional assay based on fluorescence in microtiter plate wells to allow for quantification of RBP binding. In this assay, the bacteria were mixed with RBP2, and the fluorescence of the whole well was measured. Under these conditions, most fluorescent proteins initially associated with the bacteria should remain in the samples irrespective of potential subsequent dissociation. In this assay, the RBP2 of ΦK was found to have similar affinities for the various *S. aureus* and *S. epidermidis* test strains as in the flow cytometry assay. RBP2 of ΦK adsorbed most effectively to USA300 Δ*tarM* Δ*tarS* with unglycosylated RboP-WTA but was also able to bind every other *S. aureus* strain carrying RboP-WTA, albeit with lower affinity (Figure 2E). The strong binding of ΦK to all *S. aureus* strains, including the Δ*tarM* Δ*tarS* mutant strain, can thus be explained by strong consecutive binding of the two RBPs, with the reversible RBP2 binding facilitating the binding of RBP1. While RBP1 preferentially binds to glycosylated WTA in a stable fashion, the reversibly binding RBP2 prefers unglycosylated RboP-WTA. By contrast, binding of ΦK to GroP-WTA-bearing *S. epidermidis* appears to be largely mediated by RBP1. We hypothesize that the weak contribution of RBP2 to *S. epidermidis* binding may be the reason for the comparatively low binding efficiency and, thus, reduced infection of *S. epidermidis* by ΦK (Figures S3 and S4).

WTA-dependent RBP binding patterns match phage infection behavior

Phage-*S. aureus* interactions have been studied previously using only a small set of model phages. The receptor specificity of most *S. aureus* phages with regard, for instance, to the specific WTA glycosylation pattern has remained superficially understood.[16,17,21] We assessed one or more representative RBPs of each of the RBP clusters (Figure 1, blue-marked RBPs) for their WTA-binding properties to elucidate the molecular basis of *S. aureus* phage recognition.

The RBP1 cluster of rosenblumviruses, which lack an RBP2 gene, can be separated into two subclusters. ΦP68, a representative of the first subcluster, exclusively infects *S. aureus* Δ*tarM* mutants, which carry β-1,4-GlcNAc glycosylated WTA, as shown previously.[21] Similarly, the RBP1 of ΦP68 (Figures 3A and 3B) only showed strong adsorption to the β-1,4-GlcNAc-carrying Δ*tarM* mutant of USA300 JE2 and minimal binding to the unglycosylated Δ*tarM* Δ*tarS* strain, although this low binding did not correspond to the adsorption or infection pattern of the phage itself (Figure 3A). Accordingly, rosenblumviruses of subcluster a could only bind to β-1,4-GlcNAc carrying *S. aureus*.

Since *Rosenblumvirus* ΦCSA13 (reference RBP1 for subcluster b) was not available to be tested for its host range, published data of the closely related phage ΦS24-1 was used for whole-phage assays, since its RBP1 is also found in the same subcluster (92.5% sequence identity to the RBP1 of ΦCSA13). In accordance with published data, the RBP1 of ΦCSA13 showed the same binding pattern as the phage, with no binding to the Δ*tagO* mutant and some, albeit very low, binding to the unglycosylated Δ*tarM* Δ*tarS* mutant. These results show that rosenblumviruses of subcluster b are able to bind α- and β-1,4-GlcNAc WTA carrying *S. aureus* and, to a very low extent, even unglycosylated RboP-WTA.

We observed that *S. aureus* Δ*tarM* Δ*tarS* was bound strongly by the Azeredovirinae Φ11-RBP1, whereas the WTA-deficient strain could not be bound (Figure 3A). However, Φ11 itself was neither able to bind to nor infect the Δ*tarM* Δ*tarS* mutant (Figures 3A and S3). To analyze the discrepancy between binding capacity of the phage and its RBP, the putative RBP2 of Φ11 was investigated. RBP2 of Φ11 did not show any binding in flow cytometry, which could be due to the short-lived binding followed by quick dissociation, as also observed for other RBP2 proteins. Because of this instability, the abovementioned microtiter fluorescence reader assay was used to analyze binding of Φ11-RBP2. In contrast to the negative control, the Δ*tagO* mutant, we detected strong binding of Φ11-RBP2 to USA300 WT (mainly α-1,4-GlcNAc), Δ*tarM* (β-1,4-GlcNAc), and Δ*tarS* (α-1,4-GlcNAc), while we observed no binding to the unglycosylated *S. aureus* Δ*tarM* Δ*tarS* mutant or to the GroP-WTA carrying *S. epidermidis* 1457 (Figures 3 and S3). Thus, Azeredovirinae likely depend on both RBP1 and RBP2 and can therefore bind and infect *S. aureus* with either α- or β-1,4-GlcNAc RboP-WTA.

RBP1 proteins of *Triavirus* Φ3A and Φ47 have 99.8% and 100% identity, respectively, to RBP1 of the *Triavirus* type phage

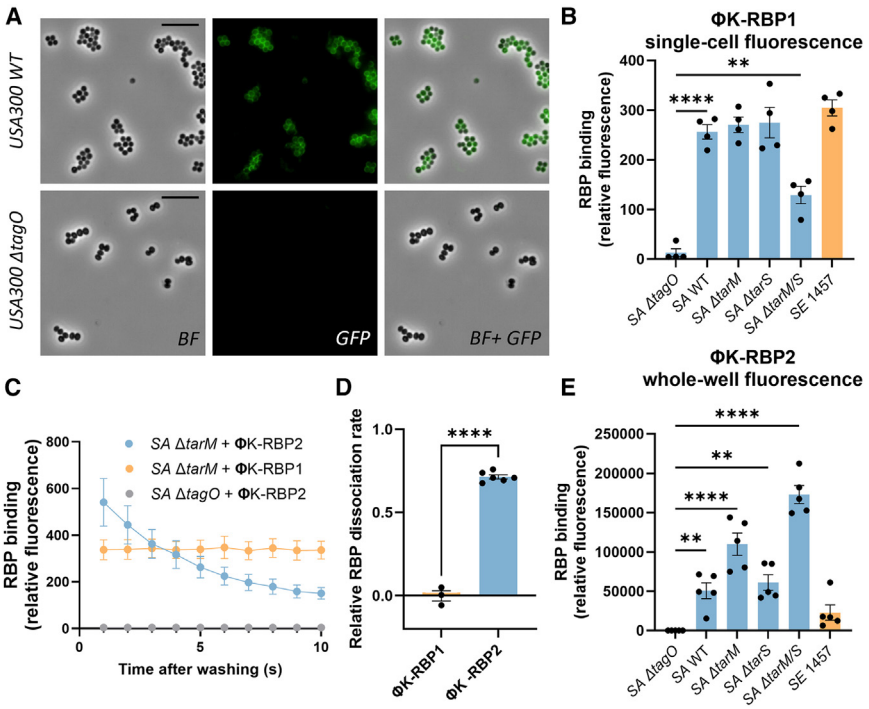


Figure 2. ΦK carries two RBPs for reversible and irreversible binding, both shaping the infection behavior of the phage

(A and B) Fluorescent GFP-RBP constructs were incubated with various *S. aureus* USA300 JE2 WTA glycosyltransferase mutants as well as a WTA-deficient control mutant (*S. aureus* USA300 JE2 Δ*tagO*) and GroP-WTA carrying *S. epidermidis* 1457. Fluorescence of the ΦK-RBP1-bound bacteria was assessed via microscopy (A; scale bars, 10 μm) and flow cytometry (B). Quantification was achieved by measuring the median fluorescence intensity.

(C and E) RBP2 binding of ΦK was measured over time via flow cytometry (C) and in a fluorescence reader in 96-well format (E).

(D) The dissociation rate of the RBPs was calculated as relative decrease in fluorescence between 1 and 10 s after washing.

Statistical analysis was done via ordinary one-way ANOVA with multiple comparisons between the WTA-negative *S. aureus* Δ*tagO* strain and the differently glycosylated strains (Table 1) for (B) $n = 4$ and (E) $n = 5$. A t test was performed for (D) with $n = 3$ for RBP1 and $n = 6$ for RBP2. Data are represented as mean ± SEM. $^{**}p < 0.01$, $^{****}p < 0.0001$.

Φ12. The binding of the Φ12-RBP1 did not match the behavior of the triaviruses (Figure 3A). Similar to Azeredovirinae, an RBP2 can be found in all triaviruses and may account for the differences in binding by RBP1 proteins and the full phages (Figure 3A). The phages of this cluster were found to infect *S. aureus* irrespective of glycosylation, although the plating efficiency was slightly reduced for the unglycosylated Δ*tarM* Δ*tarS* mutant compared to the WT (Figure S3). While the RBP1 of Φ12 only bound to the β-1,4-GlcNAc carrying Δ*tarM* and unglycosylated Δ*tarM* Δ*tarS* mutants, RBP2 strongly bound to the α-1,4-GlcNAc carrying the WT and Δ*tarS* mutant. This pattern could indicate that both RBPs of Φ12 contribute equally to the binding of the phage. Ultimately, Azeredovirinae are able to infect *S. aureus* irrespective of the presence or absence of WTA glycosylation, although with varying efficiency.

Bronfenbrennervirinae Φ13 was found to bind strongly to the *S. aureus* WT as well as to its glycosyltransferase single mutants and showed weaker, albeit still robust, binding to the unglycosylated Δ*tarM* Δ*tarS* mutant. As a typical representative of the Bronfenbrennervirinae, Φ13 encodes only one putative RBP. In adsorption experiments, Φ13 bound to *S. aureus* test strains in a similar way as its RBP1, although binding differences between Δ*tarM* Δ*tarS* and the other strains were only minor, which was probably due to the overall high adsorption capacity of Φ13-RBP1 (Figures 3 and S3). In the spot assay, we observed lysis zones on *S. aureus* RN4220 WT, RN4220 Δ*tarM*, and RN4220 Δ*tarS* and a more opaque zone on RN4220 Δ*tarM* Δ*tarS*. This observation indicates that infection of Φ13 only requires binding of its single RBP1 and that this RBP can bind to RboP-WTA irrespective of its glycosylation pattern. Bronfenbrennervirinae thus bind *S. aureus* with α- and β-1,4-GlcNAc as well as unglycosylated RboP-WTA.

As described above, Twortvirinae ΦK carries two RBPs, both of which are able to bind to all strains of *S. aureus* except the WTA-negative Δ*tagO* mutant. The GroP-WTA carrying *S. epidermidis* 1457 was only bound by ΦK-RBP1 but not by ΦK-RBP2. The Twortvirinae ΦStab20-RBP1, on the other hand, only bound to *tarM*-deficient *S. aureus* (β-1,4-GlcNAc), as shown in the binding assay (Figure 3A). In spot assays, ΦStab20 displayed the same pattern (Figure S3). This behavior is likely due to differences in the amino acid sequences of the RBP1 proteins of ΦK and ΦStab20. Accordingly, the RBP1 proteins of ΦK and ΦStab20 belong to different subclusters (Figure 1A, subclusters f and g). Across the different Twortvirinae RBP1 subclusters, we observed a variety of RBP binding patterns that correlated with the behavior of the respective phages (Figure 3A). Nevertheless, for some Twortvirinae, such as ΦPG-2021_10, the binding behavior of RBP2 did not match the phage infection pattern. Here, only binding to *S. aureus* Δ*tarM* Δ*tarS* with unglycosylated WTA could be observed, while RBP1 as well as the phage itself bound only to the *tarM*-deficient mutant. These results show that Twortvirinae binding differs between subclusters and can range from binding to every tested WTA type (ΦK) to only β-1,4-GlcNAc-modified and GroP-WTA (ΦStab20).

Overall, the binding capacities of the various RBPs largely matched the binding and infection behavior of the corresponding phages.

PhARIS detects phage RBPs and predicts their binding specificity

To enable quick detection of phage RBPs in genomes of *S. aureus* phages and prediction of the WTA-dependent host

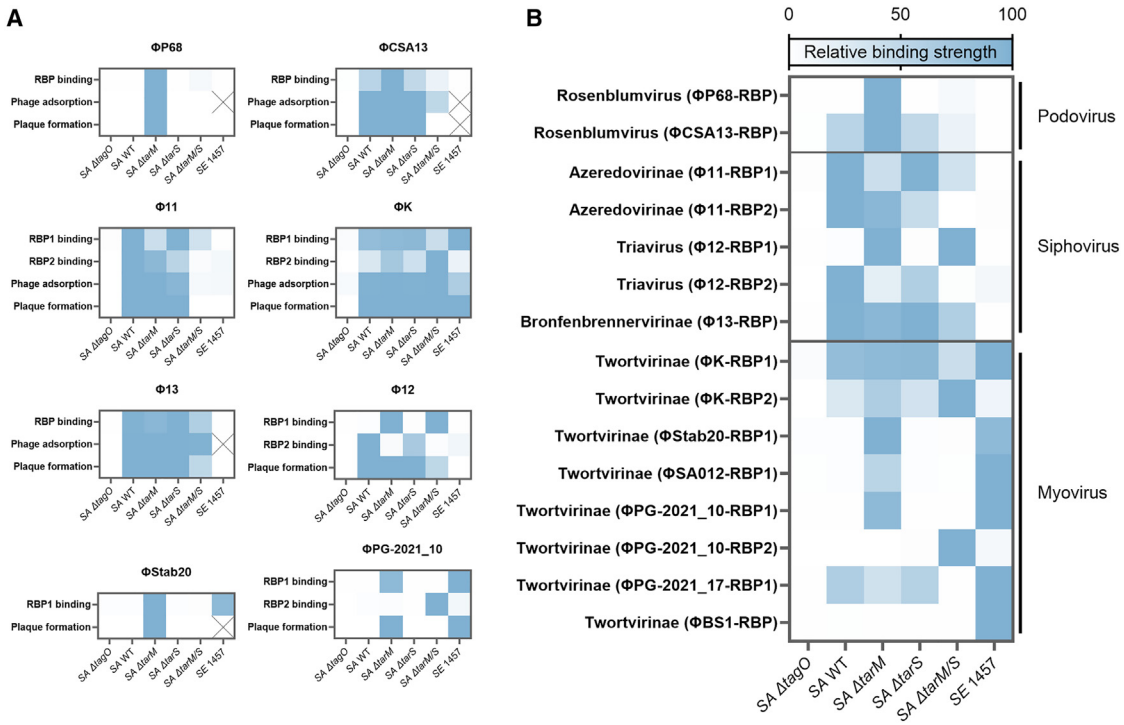


Figure 3. RBP binding matches the infection behavior of phages

(A) Phage adsorption and plaque formation of various phages in comparison with RBP binding. The host range of *Rosenblumvirus* ΦP68 and ΦCSA13 as well as Bronfenbrennervirinae Φ13 can be explained by one RBP alone. The Azeredovirinae Φ11 and Φ12, as well as the Twortvirinae ΦK, ΦPG-2021_10, and ΦStab20 (same RBP2 subcluster as ΦPG-2021_10) carry a second RBP (RBP2) that contributes to the phage host range.

(B) WTA glycosylation-dependent binding pattern of *S. aureus* phage RBPs. Distinct RBP subclusters show different WTA binding specificities. None of the RBPs could bind to the WTA-deficient Δ*tagO*, and only Twortvirinae RBPs could bind to both the GroP-WTA of *S. epidermidis* as well as the RboP-WTA of *S. aureus*. For quantification of RBP adsorption, values were normalized to the highest fluorescence of each RBP. Data are represented as mean. Plaque formation was classified into strong lysis (100%, blue), weak lysis (50%, light blue), and no lysis (0%, white). Quantitative data can be found in Figures S4 and S5.

range of the phage, we developed the PhARIS detection tool, using the data obtained as outlined in the previous paragraphs (available via https://github.com/JKrusche1/PhARIS). This computational tool allows users to upload even uncharacterized *S. aureus* phage genomes and predict their potential RBPs, the most similar RBP in the RBP cluster collection, and the likely binding specificity of the phage based on similarity to already characterized RBPs. PhARIS' functionality is based on the following features. The program finds RBP1 and RBP2 of an input phage genome by comparing protein sequences with those of 12 reference RBPs (eight RBP1 and four RBP2). The presence of likely RBP1 or RBP2 candidates and the closest relative among the 12 reference RBPs is elucidated by BLAST protein analysis. The program output includes the sequence of phage RBPs, the RBP cluster to which it was assigned, and the similarity to the reference RBP in percent. Moreover, the most similar RBP from the respective cluster, the degree of similarity, and the binding pattern of the reference RBP, which is projected to be the same as that of the input RBP, are indicated.

PhARIS was validated by analyzing the genomes and putative RBP sequences of six phages with previously unknown host range, followed by detection of WTA binding specificities of the phages using our set of *S. aureus* and *S. epidermidis* WT and mutant strains. Four new *S. aureus*-specific phages were isolated from wastewater, and two new phages were obtained from collaborators. Of the four wastewater-derived phages, two each were found to be Φ11-related Azeredovirinae and Φ12-related triaviruses. The other two phages were assigned to the phage subfamily Twortvirinae (ΦK-like) and genus *Rosenblumvirus* (ΦP68-like) (Table S5). PhARIS predicted RBP1 for all phages and RBP2 for all phages except *Rosenblumvirus*. All six phages exhibited the predicted binding behavior to the *S. aureus* and *S. epidermidis* strain panel in phage spot assays, reflecting the binding preferences of the corresponding reference phages (Figure S3; Table S5). Twortvirinae ΦMallokai, recently used successfully in patients for the therapy of MRSA infections,[40,41] was predicted to bind to *S. aureus* irrespective of WTA glycosylation and to *S. epidermidis* 1457 similarly as phage ΦK. ΦMallokai indeed bound to *S. aureus* JE2 and *S. epidermidis* 1457 but did not form plaques on *S. epidermidis* 1457 (Figure S3). An intracellular phage immunity mechanism of *S. epidermidis* 1457 protecting from ΦMallokai may be the reason for this discrepancy.

Homology predictions generated with PhARIS have also been used successfully in a previous publication focusing on Twortvirinae ΦStab20, albeit without naming the PhARIS tool or

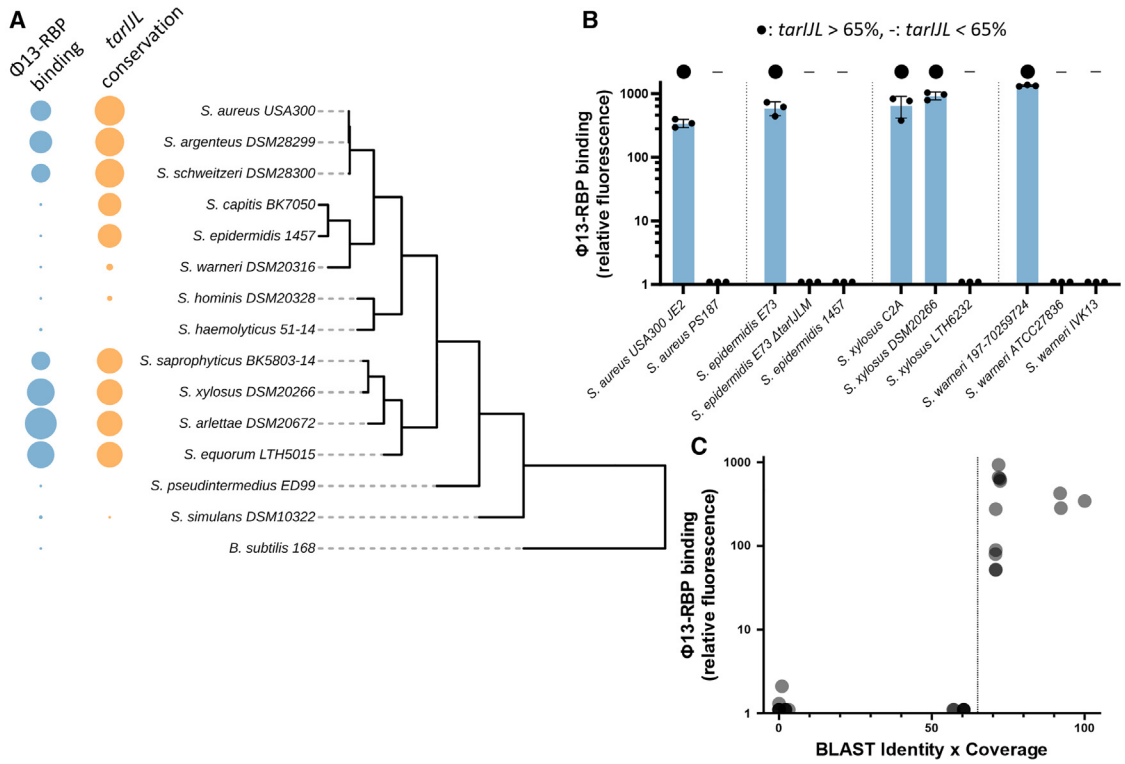


Figure 4. Φ13-RBP is a viable tool to detect RboP-WTA in a wide range of staphylococci

(A) Putative presence of RboP-WTA as indicated by Φ13-RBP binding and *tarlJL* cluster conservation (BLAST identity × coverage) for different *Staphylococcus* species and isolates. The phylogenetic tree is based on the 16S rDNA sequence.

(B) Fluorescence of different strains of *S. aureus*, *S. epidermidis*, *S. xylosus*, and *S. warneri* after co-incubation with GFP-coupled Φ13-RBP after subtracting the background fluorescence of untreated bacteria. When lower than 1.1, the resulting value was set to 1.1 for visualization purposes. Presence of *tarlJL* with above 65% identity × coverage value is indicated with ● and absence by −. Data are represented as mean ± SEM; $n = 3$.

(C) Fluorescence of GFP-coupled Φ13-RBP coincided with the presence of *tarlJL* in the genome of the corresponding bacteria. *S. aureus* USA300 JE2 *tarlJL* was used as query, and the resulting identity percentage was multiplied with the coverage percentage. Each dot represents one tested bacterial strain. Strains with identity × coverage scores higher than 65% (marked by the horizontal line) were always bound by the Φ13-RBP, while strains with lower scores did not appear to have any RboP-WTA. Full data can be found in Table S4.

describing its features.[23] Several new phages were identified with PhARIS that were projected to have a similar host range as ΦStab20. While the phages ΦvB_SauM_EW18, EW26, EW29, and EW72 did not have the highest overall similarity to ΦStab20 compared with other phages, their RBPs were found to be very similar to that of ΦStab20 (unpublished data). A host range similar to ΦStab20 was later verified by established assays and was used to bolster the findings of Yang et al.,[23] thereby underlining the predictive power of the tool.

Φ13-RBP can be used to rapidly detect RboP-WTA in a strain-specific manner

The RBP of Φ13, for which no second RBP was found, showed high affinity for GlcNAc-glycosylated RboP-WTA and moderate affinity for unglycosylated RboP-WTA, while there was no affinity for GroP-WTA (Figure 3). Based on this finding, we used the RBP of Φ13 to assess the presence of RboP in staphylococcal species with unknown WTA composition (Figure 4A). RboP-WTA is typically found in almost every *S. aureus* strain, and it can contribute to the pathogenicity of rare RboP-WTA carrying NASs, as demonstrated recently for *S. epidermidis* E73.[26] Because of the important role of RboP-WTA for virulence and binding by phages, which may facilitate horizontal gene transfer,[26] it is vital to detect RboP in prevalent and emerging clones of staphylococcal pathogens and commensals.

Φ13-RBP binding was detected in close relatives of *S. aureus*, such as *Staphylococcus schweitzeri* and *Staphylococcus argenteus*, as well as the more distantly related *Staphylococcus equorum*, *Staphylococcus arlettae*, and *Staphylococcus saprophyticus* and in most strains of *Staphylococcus xylosus,* suggesting that these strains produce RboP-WTA. In contrast, RboP-WTA dependent Φ13-RBP binding was not observed in strains of various other species, including *Staphylococcus capitis*, *Staphylococcus hominis*, *Staphylococcus simulans*, *Staphylococcus pseudintermedius*, *Staphylococcus haemolyticus*, and *Bacillus subtilis* (Figures 4A and S6B).

Strain-specific differences in Φ13-RBP binding could be detected for a variety of NAS species, suggesting that they differ in the presence of RboP-WTA. Most *S. epidermidis* carry GroP-WTA decorated with glucose residues,[25] but some clonal groups, such as *S. epidermidis* ST10, ST23, and ST85, have recently been found to produce both GroP and RboP-WTA.[26]

We could confirm this finding using the Φ13-RBP, as it bound to *S. epidermidis* E73 (sequence type 23) while showing no adsorption to *S. epidermidis* E73 Δ*tarIJLM*, whose genes required for RboP-WTA production had been deleted, or *S. epidermidis* 1457 (sequence type 86; Figure 4B). For many strains of *S. xylosus*, we found evidence of RboP-WTA in the cell wall, but strain LTH6232 was not bound by Φ13-RBP (Figure 4B). Conversely, most of the tested *S. warneri* strains did not seem to carry RboP-WTA, except for the clinical strain 197-70259724 (Figure 4B). To confirm the presence or absence of RboP-WTA, and correlate Φ13-RBP binding with the presence of RboP-WTA, we isolated the WTA of *S. xylosus* C2A, DSM20266, and LTH6232 as well as of *S. warneri* 197-70259724, ATCC27836, and IVK13, which was then analyzed for the presence of RboP and GroP constituents via mass spectrometry (Figure S6A). This experiment revealed that the strains that were bound by Φ13-RBP (*S. xylosus* C2A, *S. xylosus* DSM20266, and *S. warneri* 197-70259724) indeed possess RboP-WTA in addition to GroP-WTA, while the strains that were projected not to carry any RboP-WTA (*S. xylosus* LTH6232, *S. warneri* ATCC27836, and *S. warneri* IVK13) only showed GroP-WTA in the analysis. These results demonstrate that the GFP-fused Φ13-RBP protein can be helpful for quickly and reliably detecting RboP-WTA in bacterial isolates.

The observed strain-specific differences prompted further investigation of the correlation between Φ13-RBP binding and presence of the *tarIJL* gene cluster. The *tarIJL* genes are necessary for the production of RboP-WTA in *S. aureus*[42] and in *S. epidermidis* E73.[26] BLAST analysis was used to identify *S. aureus* USA300 JE2 *tarIJL*-homologous genes in the genomes of all tested strains (Figure 4C). Many strains carried parts of the *tarIJL* cluster in their genome, but in some of them, including *S. epidermidis* 1457 and *S. capitis*, the *tarL* gene was substantially different from that of *S. aureus*, resulting in reduced alignment coverage (around 80%). This difference coincided with the absence of Φ13-RBP binding and, presumably, RboP-WTA. Indeed, the *tarIJL1* gene cluster of *S. epidermidis* 1457 has been found recently to be nonfunctional.[26] In total, Φ13-RBP binding could be detected at identity × coverage values of above 65%, while genomes with lower *tarIJL* similarity were not able to bind Φ13-RBP (Figure 4C, bb: ● for *tarIJL* > 65%, – for *tarIJL* < 65%). The strong correlation between Φ13-RBP binding and presence of a well-conserved *tarIJL* cluster suggests that it is possible to estimate whether a specific *Staphylococcus* strain carries RboP-WTA via evaluation of the identity × coverage score related to *S. aureus tarIJL*. Such a correlation could only be demonstrated for the genus *Staphylococcus* but not for other genera, as, for example, the RboP-WTA carrying *Bacillus spizizenii* W23[43] was bound by the Φ13-RBP (Figure S6B) and carries a *tarIJL*-related gene cluster, while the identity × coverage score was only 34.6%.

DISCUSSION

This study identifies many previously unknown RBPs of *S. aureus* phages and correlates the capacities of fluorescently labeled RBPs to bind to different WTA types with the host range of the corresponding phage. All 15 analyzed RBPs of *S. aureus* phages bound to WTA, thereby confirming the notion that WTA is the dominant, if not the only, receptor for typical *S. aureus* phages. The various RBP clusters have distinct preferences for binding to specific WTA backbone or glycosylation types. Our data confirm that the RBPs of most rosenblumviruses cannot bind to TarM-modified α-1,4-GlcNAc RboP-WTA,[21,23] and they reveal the unexpected fact that some Twortvirinae RBPs are also prevented from binding when the WTA is glycosylated via TarM. Moreover, the RBPs from Twortvirinae that infect both *S. aureus* and NAS species also bound equally to RboP-WTA and GroP-WTA. The high intra-cluster similarity may reflect the conserved architecture of the specific WTA glycotypes. While the RBPs of phages that bind protein receptors might continuously evolve in response to the hosts' attempts to escape phage binding by mutating the receptor protein structure (for instance, in the case of the *E. coli* phage receptor LamB[44]), the *S. aureus* WTA structure cannot be altered by point mutations.[19] *S. aureus* could, in principle, inactivate its WTA glycosyltransferases, but such mutants have never been reported, probably as a consequence of severe fitness losses of *S. aureus* strains with non-glycosylated WTA, which cannot bind to epithelial cells anymore.[45]

We also found that *S. aureus* phages from the groups of Azeredovirinae, *Triavirus*, and Twortvirinae carry two different RBPs that bind WTA with different affinities. RBP2 is probably responsible for initial, reversible attachment, while RBP1 subsequently tethers the phage more stably to the host cells. Phages specific for other bacterial genera have been reported to encode more than one RBP to detect and bind their host. For example, the *c2* phages of *Lactococcus lactis* first bind to carbohydrates on the cell surface and then to a membrane-attached protein.[46] *S. aureus* phages, however, even when carrying two RBPs, only utilize WTA as binding epitope. The exact mechanisms of initial and tight phage binding to WTA are still unclear and necessitate further research.

The RBPs from Twortvirinae were much more heterogeneous than those of other *S. aureus* phage families, possibly because many Twortvirinae have very broad host ranges and may have evolved in different habitats on different groups of host species, such as NASs. In the case of Twortvirinae ΦSA012, minor changes in RBP amino acid sequence have been shown to cause substantial changes in RBP binding specificity and phage host range.[11] Such functional variations could explain the strong binding differences between Twortvirinae RBP subclusters and should prompt further research of the variation of the RBP sequences. The WTA binding sites of these RBPs are still unknown and could be elucidated by co-crystallization of the different RBPs with synthetic RboP-WTA fragments, which have been used successfully for the structures of WTA-antibody and WTA-glycosyltransferase complexes.[16,47] The exact stoichiometry and position of RBP1 and RBP2 proteins in the baseplates or tail fibers of the various phage families also remain to be analyzed.

The functional characterization of representatives of the various RBP clusters provides the basis for the prediction of RBP genes in phage genomes and of host ranges of the corresponding phage. Our PhARIS tool facilitates the characterization of phages regarding the identity and specificity of their RBPs and prediction of potentially susceptible *S. aureus* lineages. PhARIS will therefore be very helpful for the identification of promising

phages for phage therapies against *S. aureus*. Using this approach, we were able to find phages with a similar host range as Twortvirinae ΦStab20, and we could show recently that quorum sensing-dependent downregulation of the WTA glycosyltransferase gene *tarM* can impact infection of *S. aureus* by multiple different Twortvirinae.[23] In the long term, PhARIS can be upgraded to also cover phages infecting NASs.

Φ13-RBP was not only helpful for the analysis of the receptor specificities of Φ13-related Bronfenbrennervirinae but also as a tool to identify NAS isolates with RboP-WTA. Φ13-RBP binding assays, combined with genome mining to detect the presence of the *tarIJL* gene cluster responsible for RboP-WTA synthesis and with WTA compositional analysis, demonstrated that Φ13-RBP easily detects the presence of RboP-WTA. In contrast to most *S. warneri* species, which did not carry *tarIJL*, the clinical isolate *S. warneri* 197-70259724 was bound by the Φ13-RBP, encoded *tarIJL*, and produced RboP-WTA. This finding may help us to understand the virulence properties of this isolate, as RboP-WTA has been associated with NAS virulence by enhanced binding to endothelial cells.[26] In this context, the Φ13-RBP could be a valuable tool for finding RboP-WTA-carrying species and strains even in distantly related Firmicutes, such as *B. spizizenii* or *Listeria monocytogenes* in a high-throughput fashion. Additionally, Φ13-RBP could be useful for targeted drug delivery to RboP-WTA expressing pathogens (for instance, with Φ13-RBP coupled to antibacterial lysins), which might permit the specific targeting of RboP-WTA carrying bacteria with high virulence potential.[48] In this way, the binding affinity of antibacterial lysins could be improved, increasing their efficacy and reducing the required dose. The rapid and reliable detection of RboP-WTA in NASs is also of relevance for understanding routes of horizontal transfer of antimicrobial resistance and virulence genes between NASs and *S. aureus* by RboP-WTA-binding transducing phages.

Limitations of the study

Our study represents an important advance for predicting the suitability of phages for interventions against specific *S. aureus* strains, based on phage genome sequences, but it has its limitations. PhARIS can currently only analyze *S. aureus*-infecting phages, while analysis of phages specific for NASs or even more distantly related bacterial species will need extensive upgrades of the tool. While PhARIS can predict the binding capacities of a specific phage, the infection capacity can also be governed by additional bacterial phage immunity mechanisms. Moreover, even slight changes in RBP sequence could lead to altered binding specificities that PhARIS could not predict. Nevertheless, none of the six tested RBPs diverted from the predicted binding specificity, indicating that such specificity-changing sequence alterations may be rare.

RESOURCE AVAILABILITY

Lead contact

Requests for further information, resources, and reagents should be directed to and will be fulfilled by the lead contact, Andreas Peschel (andreas.peschel@uni-tuebingen.de).

Materials availability

All materials generated in this study are available from the lead contact upon request.

Data and code availability

- Phage genome data have been deposited at the NCBI as GenBank: PV069679-PV069681 and are publicly available as of the date of publication. All data reported in this paper will be shared by the lead contact upon request.
- All original code has been deposited on GitHub and is publicly available at https://github.com/JKrusche1/PhARIS as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, J.K., C.B., and A.P.; methodology, J.K., C.B., E.L., D.G., C.W., and A.P.; investigation, J.K., C.B., E.D., E.L., C.M., J.M., H.O., and S.W.; formal analysis, J.K., C.B., and J.M.; visualization, J.K., D.G., and A.P.; writing – original draft, J.K.; writing – review & editing, J.K., C.B., E.L., D.G., S.W., and A.P.; supervision, A.P., funding acquisition, A.P.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Lee, A.S., de Lencastre, H., Garau, J., Kluytmans, J., Malhotra-Kumar, S., Peschel, A., and Harbarth, S. (2018). Methicillin-resistant Staphylococcus aureus. Nat. Rev. Dis. Primers *4*, 18033. https://doi.org/10.1038/nrdp.2018.33.
2. GBD 2019 Antimicrobial Resistance Collaborators (2022). Global mortality associated with 33 bacterial pathogens in 2019: a systematic analysis for the Global Burden of Disease Study 2019. Lancet *400*, 2221–2248. https://doi.org/10.1016/S0140-6736(22)02185-7.
3. Antimicrobial Resistance Collaborators (2022). Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet *399*, 629–655. https://doi.org/10.1016/S0140-6736(21)02724-0.
4. Pirnay, J.P., Djebara, S., Steurs, G., Griselain, J., Cochez, C., De Soir, S., Glonti, T., Spiessens, A., Vanden Berghe, E., Green, S., et al. (2024). Personalized bacteriophage therapy outcomes for 100 consecutive cases: a multicentre, multinational, retrospective observational study. Nat. Microbiol. *9*, 1434–1453. https://doi.org/10.1038/s41564-024-01705-x.
5. Mu, A., McDonald, D., Jarmusch, A.K., Martino, C., Brennan, C., Bryant, M., Humphrey, G.C., Toronczak, J., Schwartz, T., Nguyen, D., et al. (2021). Assessment of the microbiome during bacteriophage therapy in combination with systemic antibiotics to treat a case of staphylococcal device infection. Microbiome *9*, 92. https://doi.org/10.1186/s40168-021-01026-9.
6. Bertozzi Silva, J., Storms, Z., and Sauvageau, D. (2016). Host receptors for bacteriophage adsorption. FEMS Microbiol. Lett. *363*, fnw002. https://doi.org/10.1093/femsle/fnw002.
7. Garen, A., and Puck, T.T. (1951). The first two steps of the invasion of host cells by bacterial viruses. II. J. Exp. Med. *94*, 177–189. https://doi.org/10.1084/jem.94.3.177.
8. Koc, C., Xia, G., Kuhner, P., Spinelli, S., Roussel, A., Cambillau, C., and Stehle, T. (2016). Structure of the host-recognition device of Staphylococcus aureus phage varphi11. Sci. Rep. *6*, 27581. https://doi.org/10.1038/srep27581.
9. Kizziah, J.L., Manning, K.A., Dearborn, A.D., and Dokland, T. (2020). Structure of the host cell recognition and penetration machinery of a Staphylococcus aureus bacteriophage. PLoS Pathog. *16*, e1008314. https://doi.org/10.1371/journal.ppat.1008314.
10. Hrebik, D., Stverakova, D., Skubnik, K., Fuzik, T., Pantucek, R., and Plevka, P. (2019). Structure and genome ejection mechanism of Staphylococcus aureus phage P68. Sci. Adv. *5*, eaaw7414. https://doi.org/10.1126/sciadv.aaw7414.
11. Takeuchi, I., Osada, K., Azam, A.H., Asakawa, H., Miyanaga, K., and Tanji, Y. (2016). The Presence of Two Receptor-Binding Proteins Contributes to the Wide Host Range of Staphylococcal Twort-Like Phages. Appl. Environ. Microbiol. *82*, 5763–5774. https://doi.org/10.1128/AEM.01385-16.
12. Leprince, A., and Mahillon, J. (2023). Phage Adsorption to Gram-Positive Bacteria. Viruses *15*, 196. https://doi.org/10.3390/v15010196.
13. Weidenmaier, C., and Peschel, A. (2008). Teichoic acids and related cell-wall glycopolymers in Gram-positive physiology and host interactions. Nat. Rev. Microbiol. *6*, 276–287. https://doi.org/10.1038/nrmicro1861.
14. Xia, G., Corrigan, R.M., Winstel, V., Goerke, C., Gründling, A., and Peschel, A. (2011). Wall teichoic Acid-dependent adsorption of staphylococcal siphovirus and myovirus. J. Bacteriol. *193*, 4006–4009. https://doi.org/10.1128/JB.01412-10.
15. van Dalen, R., Peschel, A., and van Sorge, N.M. (2020). Wall Teichoic Acid in Staphylococcus aureus Host Interaction. Trends Microbiol. *28*, 985–998. https://doi.org/10.1016/j.tim.2020.05.017.
16. Gerlach, D., Guo, Y., De Castro, C., Kim, S.H., Schlatterer, K., Xu, F.F., Pereira, C., Seeberger, P.H., Ali, S., Codée, J., et al. (2018). Methicillin-resistant Staphylococcus aureus alters cell wall glycosylation to evade immunity. Nature *563*, 705–709. https://doi.org/10.1038/s41586-018-0730-x.
17. Xia, G., Maier, L., Sanchez-Carballo, P., Li, M., Otto, M., Holst, O., and Peschel, A. (2010). Glycosylation of wall teichoic acid in Staphylococcus aureus by TarM. J. Biol. Chem. *285*, 13405–13415. https://doi.org/10.1074/jbc.M109.096172.
18. Brown, S., Xia, G., Luhachack, L.G., Campbell, J., Meredith, T.C., Chen, C., Winstel, V., Gekeler, C., Irazoqui, J.E., Peschel, A., and Walker, S. (2012). Methicillin resistance in Staphylococcus aureus requires glycosylated wall teichoic acids. Proc. Natl. Acad. Sci. USA *109*, 18909–18914. https://doi.org/10.1073/pnas.1209126109.
19. Tamminga, S.M., Volpel, S.L., Schipper, K., Stehle, T., Pannekoek, Y., and van Sorge, N.M. (2022). Genetic diversity of Staphylococcus aureus wall teichoic acid glycosyltransferases affects immune recognition. Microb. Genom. *8*, mgen000902. https://doi.org/10.1099/mgen.0.000902.
20. Gerlach, D., Sieber, R.N., Larsen, J., Krusche, J., De Castro, C., Baumann, J., Molinaro, A., and Peschel, A. (2022). Horizontal transfer and phylogenetic distribution of the immune evasion factor tarP. Front. Microbiol. *13*, 951333. https://doi.org/10.3389/fmicb.2022.951333.
21. Li, X., Gerlach, D., Du, X., Larsen, J., Stegger, M., Kühner, P., Peschel, A., Xia, G., and Winstel, V. (2015). An accessory wall teichoic acid glycosyltransferase protects Staphylococcus aureus from the lytic activity of Podoviridae. Sci. Rep. *5*, 17219. https://doi.org/10.1038/srep17219.
22. Winstel, V., Sanchez-Carballo, P., Holst, O., Xia, G., and Peschel, A. (2014). Biosynthesis of the unique wall teichoic acid of Staphylococcus aureus lineage ST395. mBio *5*, e00869. https://doi.org/10.1128/mBio.00869-14.
23. Yang, J., Bowring, J.Z., Krusche, J., Lehmann, E., Bejder, B.S., Silva, S.F., Bojer, M.S., Grunert, T., Peschel, A., and Ingmer, H. (2023). Cross-species communication via agr controls phage susceptibility in Staphylococcus aureus. Cell Rep. *42*, 113154. https://doi.org/10.1016/j.celrep.2023.113154.
24. Endl, J., Seidl, H.P., Fiedler, F., and Schleifer, K.H. (1983). Chemical composition and structure of cell wall teichoic acids of staphylococci. Arch. Microbiol. *135*, 215–223. https://doi.org/10.1007/BF00414483.
25. Beck, C., Krusche, J., Notaro, A., Walter, A., Kränkel, L., Vollert, A., Stemmler, R., Wittmann, J., Schaller, M., Slavetinsky, C., et al. (2024). Wall teichoic acid substitution with glucose governs phage susceptibility of Staphylococcus epidermidis. mBio *15*, e0199023. https://doi.org/10.1128/mbio.01990-23.
26. Du, X., Larsen, J., Li, M., Walter, A., Slavetinsky, C., Both, A., Sanchez Carballo, P.M., Stegger, M., Lehmann, E., Liu, Y., et al. (2021). Staphylococcus epidermidis clones express Staphylococcus aureus-type wall teichoic acid to shift from a commensal to pathogen lifestyle. Nat. Microbiol. *6*, 757–768. https://doi.org/10.1038/s41564-021-00913-z.
27. Uchiyama, J., Taniguchi, M., Kurokawa, K., Takemura-Uchiyama, I., Ujihara, T., Shimakura, H., Sakaguchi, Y., Murakami, H., Sakaguchi, M., and Matsuzaki, S. (2017). Adsorption of Staphylococcus viruses S13' and S24-1 on Staphylococcus aureus strains with different glycosidic linkage patterns of wall teichoic acids. J. Gen. Virol. *98*, 2171–2180. https://doi.org/10.1099/jgv.0.000865.

28. Jumper, J., Evans, R., Pritzel, A., Green, T., Figurnov, M., Ronneberger, O., Tunyasuvunakool, K., Bates, R., Žídek, A., Potapenko, A., et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature *596*, 583–589. https://doi.org/10.1038/s41586-021-03819-2.

29. Soding, J., Biegert, A., and Lupas, A.N. (2005). The HHpred interactive server for protein homology detection and structure prediction. Nucleic Acids Res. *33*, W244–W248. https://doi.org/10.1093/nar/gki408.

30. Hatfull, G.F., and Hendrix, R.W. (2011). Bacteriophages and their genomes. Curr. Opin. Virol. *1*, 298–303. https://doi.org/10.1016/j.coviro.2011.06.009.

31. Cook, R., Brown, N., Redgwell, T., Rihtman, B., Barnes, M., Clokie, M., Stekel, D.J., Hobman, J., Jones, M.A., and Millard, A. (2021). INfrastructure for a PHAge REference Database: Identification of Large-Scale Biases in the Current Collection of Cultured Phage Genomes. Phage (New Rochelle) *2*, 214–223. https://doi.org/10.1089/phage.2021.0007.

32. Sievers, F., Wilm, A., Dineen, D., Gibson, T.J., Karplus, K., Li, W., Lopez, R., McWilliam, H., Remmert, M., Söding, J., et al. (2011). Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. Mol. Syst. Biol. *7*, 539. https://doi.org/10.1038/msb.2011.75.

33. Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A.J., Bambrick, J., et al. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature *630*, 493–500. https://doi.org/10.1038/s41586-024-07487-w.

34. Diep, B.A., Gill, S.R., Chang, R.F., Phan, T.H., Chen, J.H., Davidson, M.G., Lin, F., Lin, J., Carleton, H.A., Mongodin, E.F., et al. (2006). Complete genome sequence of USA300, an epidemic clone of community-acquired meticillin-resistant Staphylococcus aureus. Lancet *367*, 731–739. https://doi.org/10.1016/S0140-6736(06)68231-7.

35. Atshan, S.S., Hamat, R.A., Aljaberi, M.A., Chen, J.S., Huang, S.W., Lin, C.Y., Mullins, B.J., and Kicic, A. (2023). Phage Therapy as an Alternative Treatment Modality for Resistant Staphylococcus aureus Infections. Antibiotics *12*, 286. https://doi.org/10.3390/antibiotics12020286.

36. Lehman, S.M., Mearns, G., Rankin, D., Cole, R.A., Smrekar, F., Branston, S.D., and Morales, S. (2019). Design and Preclinical Development of a Phage Product for the Treatment of Antibiotic-Resistant Staphylococcus aureus Infections. Viruses *11*, 88. https://doi.org/10.3390/v11010088.

37. D'Elia, M.A., Henderson, J.A., Beveridge, T.J., Heinrichs, D.E., and Brown, E.D. (2009). The N-acetylmannosamine transferase catalyzes the first committed step of teichoic acid assembly in Bacillus subtilis and Staphylococcus aureus. J. Bacteriol. *191*, 4030–4034. https://doi.org/10.1128/JB.00611-08.

38. Weidenmaier, C., Kokai-Kun, J.F., Kristian, S.A., Chanturiya, T., Kalbacher, H., Gross, M., Nicholson, G., Neumeister, B., Mond, J.J., and Peschel, A. (2004). Role of teichoic acids in Staphylococcus aureus nasal colonization, a major risk factor in nosocomial infections. Nat. Med. *10*, 243–245. https://doi.org/10.1038/nm991.

39. Slavetinsky, J., Lehmann, E., Slavetinsky, C., Gritsch, L., van Dalen, R., Kretschmer, D., Bleul, L., Wolz, C., Weidenmaier, C., and Peschel, A. (2023). Wall Teichoic Acid Mediates Staphylococcus aureus Binding to Endothelial Cells via the Scavenger Receptor LOX-1. ACS Infect. Dis. *9*, 2133–2140. https://doi.org/10.1021/acsinfecdis.3c00252.

40. Bernabeu-Gimeno, M., Pardo-Freire, M., Chan, B.K., Turner, P.E., Gil-Brusola, A., Perez-Tarazona, S., Carrasco-Hernandez, L., Quintana-Gallego, E., and Domingo-Calap, P. (2024). Neutralizing antibodies after nebulized phage therapy in cystic fibrosis patients. Médica Sur *5*, 1096–1111.e1096. https://doi.org/10.1016/j.medj.2024.05.017.

41. Doub, J.B., Ng, V.Y., Lee, M., Chi, A., Lee, A., Würstle, S., and Chan, B. (2022). Salphage: Salvage Bacteriophage Therapy for Recalcitrant MRSA Prosthetic Joint Infection. Antibiotics *11*, 616. https://doi.org/10.3390/antibiotics11050616.

42. Qian, Z., Yin, Y., Zhang, Y., Lu, L., Li, Y., and Jiang, Y. (2006). Genomic characterization of ribitol teichoic acid synthesis in Staphylococcus aureus: genes, genomic organization and gene duplication. BMC Genom. *7*, 74. https://doi.org/10.1186/1471-2164-7-74.

43. Brown, S., Meredith, T., Swoboda, J., and Walker, S. (2010). Staphylococcus aureus and Bacillus subtilis W23 make polyribitol wall teichoic acids using different enzymatic pathways. Chem. Biol. *17*, 1101–1110. https://doi.org/10.1016/j.chembiol.2010.07.017.

44. Chatterjee, S., and Rothenberg, E. (2012). Interaction of bacteriophage l with its E. coli receptor. Viruses *4*, 3162–3178. https://doi.org/10.3390/v4113162.

45. Winstel, V., Kühner, P., Salomon, F., Larsen, J., Skov, R., Hoffmann, W., Peschel, A., and Weidenmaier, C. (2015). Wall Teichoic Acid Glycosylation Governs Staphylococcus aureus Nasal Colonization. mBio *6*, e00632. https://doi.org/10.1128/mBio.00632-15.

46. Tremblay, D.M., Tegoni, M., Spinelli, S., Campanacci, V., Blangy, S., Huyghe, C., Desmyter, A., Labrie, S., Moineau, S., and Cambillau, C. (2006). Receptor-binding protein of Lactococcus lactis phages: identification and characterization of the saccharide receptor-binding site. J. Bacteriol. *188*, 2400–2410. https://doi.org/10.1128/JB.188.7.2400-2410.2006.

47. Di Carluccio, C., Soriano-Maldonado, P., Berni, F., de Haas, C.J.C., Temming, A.R., Hendriks, A., Ali, S., Molinaro, A., Silipo, A., van Sorge, N.M., et al. (2022). Antibody Recognition of Different Staphylococcus aureus Wall Teichoic Acid Glycoforms. ACS Cent. Sci. *8*, 1383–1392. https://doi.org/10.1021/acscentsci.2c00125.

48. Zampara, A., Sørensen, M.C.H., Grimon, D., Antenucci, F., Vitt, A.R., Bortolaia, V., Briers, Y., and Brøndsted, L. (2020). Exploiting phage receptor binding proteins to enable endolysins to kill Gram-negative bacteria. Sci. Rep. *10*, 12087. https://doi.org/10.1038/s41598-020-68983-3.

49. Hendriks, A., van Dalen, R., Ali, S., Gerlach, D., van der Marel, G.A., Fuchsberger, F.F., Aerts, P.C., de Haas, C.J.C., Peschel, A., Rademacher, C., et al. (2021). Impact of Glycan Linkage to Staphylococcus aureus Wall Teichoic Acid on Langerin Recognition and Langerhans Cell Activation. ACS Infect. Dis. *7*, 624–635. https://doi.org/10.1021/acsinfecdis.0c00822.

50. Gibson, D.G., Young, L., Chuang, R.Y., Venter, J.C., Hutchison, C.A., 3rd, and Smith, H.O. (2009). Enzymatic assembly of DNA molecules up to several hundred kilobases. Nat. Methods *6*, 343–345. https://doi.org/10.1038/nmeth.1318.

51. Winstel, V., Liang, C., Sanchez-Carballo, P., Steglich, M., Munar, M., Bröker, B.M., Penadés, J.R., Nübel, U., Holst, O., Dandekar, T., et al. (2013). Wall teichoic acid structure governs horizontal gene transfer between major bacterial pathogens. Nat. Commun. *4*, 2345. https://doi.org/10.1038/ncomms3345.

52. Bae, T., and Schneewind, O. (2006). Allelic replacement in Staphylococcus aureus with inducible counter-selection. Plasmid *55*, 58–63. https://doi.org/10.1016/j.plasmid.2005.05.005.

53. Cheung, G.Y.C., Duong, A.C., and Otto, M. (2012). Direct and synergistic hemolysis caused by Staphylococcus phenol-soluble modulins: implications for diagnosis and pathogenesis. Microb. Infect. *14*, 380–386. https://doi.org/10.1016/j.micinf.2011.11.013.

54. Adhikari, R.P., Arvidson, S., and Novick, R.P. (2007). A nonsense mutation in agrA accounts for the defect in agr expression and the avirulence of Staphylococcus aureus 8325-4 traP::kan. Infect. Immun. *75*, 4534–4540. https://doi.org/10.1128/iai.00679-07.

55. Kropinski, A.M., Mazzocco, A., Waddell, T.E., Lingohr, E., and Johnson, R.P. (2009). Enumeration of bacteriophages by double agar overlay plaque assay. Methods Mol. Biol. *501*, 69–76. https://doi.org/10.1007/978-1-60327-164-6_7.

56. Goller, P.C., Elsener, T., Lorge, D., Radulovic, N., Bernardi, V., Naumann, A., Amri, N., Khatchatourova, E., Coutinho, F.H., Loessner, M.J., and Gomez-Sanz, E. (2021). Multi-species host range of staphylococcal phages isolated from wastewater. Nat. Commun. *12*, 6965. https://doi.org/10.1038/s41467-021-27037-6.

57. Letunic, I., and Bork, P. (2021). Interactive Tree Of Life (iTOL) v5: an online tool for phylogenetic tree display and annotation. Nucleic Acids Res. *49*, W293–W296. https://doi.org/10.1093/nar/gkab301.

58. Altschul, S.F., Madden, T.L., Schäffer, A.A., Zhang, J., Zhang, Z., Miller, W., and Lipman, D.J. (1997). Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. Nucleic Acids Res. *25*, 3389–3402. https://doi.org/10.1093/nar/25.17.3389.

59. Madeira, F., Pearce, M., Tivey, A.R.N., Basutkar, P., Lee, J., Edbali, O., Madhusoodanan, N., Kolesnikov, A., and Lopez, R. (2022). Search and sequence analysis tools services from EMBL-EBI in 2022. Nucleic Acids Res. *50*, W276–W279. https://doi.org/10.1093/nar/gkac240.

60. Pereira, M.P., D'Elia, M.A., Troczynska, J., and Brown, E.D. (2008). Duplication of teichoic acid biosynthetic genes in Staphylococcus aureus leads to functionally redundant poly(ribitol phosphate) polymerases. J. Bacteriol. *190*, 5642–5649. https://doi.org/10.1128/JB.00526-08.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-β-GlcNAc WTA IgG	Hendriks et al.[49]	N/A
Anti-α-GlcNAc WTA IgG	Hendriks et al.[49]	N/A
F(ab')2-Goat anti-Human IgG-FITC	Sigma-Aldrich	Cat#AQ112F; RRID:AB_92761
Bacterial and virus strains		
Bacterial Strains, see Table S1	This paper	N/A
Bacteriophages, see Table S1	This paper	N/A
Chemicals, peptides, and recombinant proteins		
Recombinant RBPs, see Table S2		
Critical commercial assays		
NEBuilder HiFi DNA Assembly Master Mix	New England Biolabs	Cat#E2621S
GeneJET Plasmid-Miniprep-Kit	Thermo Scientific	Cat#K0503
Slide-A-Lyzer G3 dialysis cassettes	Thermo Scientific	Cat#A52971
Deposited data		
PhARIS (Phage Aureus Identification System)	This paper	GitHub: https://github.com/JKrusche1/PhARIS
ΦvB_Sau_FK1	This paper	GenBank: PV069679
ΦvB_Sau_SBK1	This paper	GenBank: PV069680
ΦvB_Sau_IS1	This paper	GenBank: PV069681
ΦvB_Sau_JK1	This paper	GenBank: PV069682
Oligonucleotides		
Primers and synthetic DNA, see Table S3	This paper	N/A
Recombinant DNA		
Plasmids, see Table S2	This paper	N/A
Software and algorithms		
FlowJo 10.9	BD Biosciences	https://www.flowjo.com/
Graphpad Prism 10.1.2	Dotmatics	https://www.graphpad.com/
iTOL – interactive Tree of Life	EMBL-EBI	https://itol.embl.de/
Protein BLAST	NIH	https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Proteins
Clustal Omega	EMBL-EBI	https://www.ebi.ac.uk/jdispatcher/msa/clustalo
SimplePhylogeny	EMBL-EBI	https://www.ebi.ac.uk/jdispatcher/phylogeny/simple_phylogeny
Python 3.12.0	Python Software Foundation	https://www.python.org/
The PyMOL Molecular Graphics System, Version 3.1	Schrödinger, LLC.	https://github.com/schrodinger/pymol-open-source
AlphaFold 3	Google DeepMind and Isomorphic Labs	https://alphafoldserver.com/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Bacteria and bacteriophages used in this study are described in Table S1.

METHOD DETAILS

Bacterial strains and growth conditions

Staphylococcus and *Bacillus* strains were grown at 37°C on an orbital shaker in Tryptic Soy Broth (TSB). *E. coli* strains were grown in Lysogeny Broth (LB) at 37°C. *Staphylococcus* and *Bacillus* species were grown without antibiotics, while *E. coli* overexpression strains were grown with 10 μg/mL kanamycin. All strains, phages, and the respective propagation hosts can be found in Table S1.

Phage propagation

Phages were propagated by inoculating a liquid culture of the phage-specific propagation strain to OD 0.4 (see Table S1) in 9 mL TSB +5 mM $CaCl_2$ and incubated at 37°C for 30 min. Next, 1 mL of phage lysate (titer 10^8-10^{10}) was added, and the mixture was incubated at 37°C for 4 h (podo- and siphoviruses) or 30°C for 6 h (myoviruses). Then, the lysate was centrifuged (5,000 x g for 5 min) and the supernatant was sterile-filtered (0.45 μm) and stored at 4°C.

Cloning of RBP-expressing *E. coli*

The His(6)-GFP-encoding DNA sequence was cloned in pET-28a(+) via Gibson assembly,[50] and used to transform chemically competent *E. coli* DC10B via heat shock for 30 s at 42°C. The RBP genes were either amplified by PCR from the phage genome or ordered as synthetic DNA fragments from Thermo Scientific via the Invitrogen GeneArt Synthesis Services in cases where the phage itself was unavailable as PCR template. Ordered primers and DNA fragments can be found in Table S3. The RBP-encoding fragments were then ligated in frame onto the C-terminal end of the GFP-encoding fragment in the pET-28a(+)_GFP vector via Gibson assembly and used once again to transform *E. coli* DC10B. From there, resulting plasmids were purified and transferred by heat shock into chemically competent *E. coli* BL21(DE3) for protein overexpression.

Cloning in *S. aureus*

Cloning of glycosyltransferase mutants in *S. aureus* USA300 JE2 was performed as previously described for RN4220 Δ*tarM* and RN4220 Δ*tarS*.[51] The marker-less deletion mutants of the glycosyltransferase genes *tarM* and *tarS*, as well as a Δ*tarM* Δ*tarS* double mutant, were originally generated by Gibson cloning of the two flanking regions of the respective genes in the mutagenesis vector pBASE6 or pKOR1. These original mutagenesis vectors were transferred by electroporation (pBASE6-tarM) and transduction via ϕ11 (pKOR1-TarS) into USA300 JE2, following the recombination procedure described elsewhere.[52] The mutants were confirmed genotypically by PCR and sequencing and controlled for *agr* activity and toxin production by cultivation on blood agar plates,[53,54] coagulase activity and deposition of anti-WTA-Fabs (using clones 4461 and 4497[49]). The strains, plasmids and oligonucleotides used are listed in the supplemental information in Tables S1, S2, and S3, respectively.

Overexpression and purification of phage receptor-binding proteins

E. coli BL21(DE3) carrying the respective pET-28a(+)_GFP_RBP plasmids (See Table S2) were incubated overnight in LB at 37°C shaking. Overexpression cultures were inoculated to OD_{600} 0.1 in TSB and incubated for 2–3 h at 37°C under shaking at 200 rpm, whereafter the temperature was shifted to 20°C for 15 min. Subsequently the overexpression was induced by addition of 1 μg/mL Isopropyl β-D-1-thiogalactopyranoside (IPTG) and incubated on a shaker overnight at 20°C.

The cells were collected by centrifugation (10 min at 4,000 x g), resuspended in lysis buffer (30 mM Tris-HCl pH 8.3 with 20 mM imidazole and 300 mM NaCl), treated with lysozyme, Triton X-100, and protease inhibitor tablets, and lysed by sonication. After removal of cell debris by centrifugation (17,000 x g, 10 min), the supernatant was sterile-filtered (0.22 μm), and the proteins were purified via nickel column chromatography. Smaller impurities and lysis buffer contents were removed by dialysis with Slide-A-Lyzer G3 dialysis cassettes (10K MWCO) Cat. Nr. A52971 overnight in RBP buffer (30 mM Tris-HCl pH 8.3 with 150 mM NaCl) at 4°C. Protein concentration was measured after dialysis via the Qubit Protein Assay.

Spot assay

The phage count was enumerated as plaque-forming units (PFU) per mL via agar overlay method.[55] Briefly, *S. aureus* and *S. epidermidis* cultures were inoculated from overnight cultures at an OD_{600} of 0.05 in 4 mL of 0.5% TSA soft agar (TSB with 0.5% agarose) + 5 mM $CaCl_2$. The soft agar was poured onto TSA plates. After solidification, 5 μL of dilution series of the phages were spotted onto each soft agar plate containing a different bacterial strain and incubated overnight at 37°C (sipho- and podoviruses) or 30°C (myoviruses).

Adsorption assay

The phage adsorption assay was performed as described in[14] with slight modifications. Briefly, 100 μL of phage lysate containing 3 x 10^7 or 3 x 10^8 phages was mixed with 200 μL of bacteria (OD_{600} 0.5) and incubated for 10 min at 30°C with shaking (300 rpm). Afterward, the bacteria and bound phages were removed by centrifugation at 13.000 x g for 5 min at 4°C and subsequent filtration of the supernatant (0.45 μm). The remaining phage lysates were serially diluted and then used in a spot assay for enumeration. The percentage of bound phages was calculated as ratio compared to the negative control that contained no bacteria during the initial incubation step.

RBP flow cytometry assay

Overnight cultures of the test strains were washed once in RBP buffer (30 mM Tris-HCl pH 8.3 with 150 mM NaCl) and then diluted to OD_{600} 0.4. 30 μL of bacteria were mixed with 30 μL of the purified RBPs (0.2 μM) and incubated for 8 min (Fluorescence did not increase with longer incubation time) at 20°C shaking (350 rpm). Next, the bacteria were washed by addition of 90 μL RBP buffer followed by centrifugation for 2.5 min at 8,000 x g. The bacteria were resuspended in 150 μL fresh RBP buffer. The bacterial suspension was then transferred into FACS tubes and GFP-mediated fluorescence of the cells was measured via flow cytometry in a BD FACSCalibur (FL1).

RBP fluoreader assay

Overnight cultures of the test strains were washed once in RBP buffer (30 mM Tris-HCl pH 8.3 with 150 mM NaCl) and then diluted to OD_{600} = 2. 30 μL of bacteria were mixed with 30 μL of the purified RBPs (2 μM) and incubated for 4 min (Fluorescence did not increase with longer incubation time) at 20°C with shaking (350 rpm). Next, the bacteria were washed by centrifugation for 2.5 min at 8,000 x g, whereafter the bacteria were resuspended in 150 μL RBP buffer. The fluorescence of the samples was measured in a 96-well plate in a BMG CLARIOstar (Ex: 470-15; Em: 515-20).

Fluorescence microscopy

Bacteria and phage RBPs were processed as in the RBP FACS assay. After washing, 32 μL of the bacterial suspension was transferred into μ-Slide 15 Well 3D (formerly μ-Slide Angiogenesis) ibidi Cat.No:81506 and centrifuged twice for 6 min at 600 x g. The supernatant was carefully removed, and the wells were filled with 10 μL ibidi mounting medium Cat.No:50001. The samples were analyzed using a fluorescence microscope.

Phage isolation

New bacteriophages were isolated similarly to.[56] 1 L of incoming wastewater was collected and centrifuged at 10,000 x g for 10 min. Afterward, the supernatant was filtered (0,45 μm). 4 x 40 mL wastewater was then mixed with 10 mL 5x TSB each, supplemented with 5 mM $CaCl_2$ (final concentration), and then inoculated with 40 μL overnight culture of *S. aureus* RN4220 WT (2x) or RN4220 Δ*tarM* (2x). These were then incubated overnight at 30°C and 37°C (one for each strain). The next day, the cultures were centrifuged at 5,000 x g for 5 min and filtered (0,45 μm). The enrichment process was repeated one more time. After this, 10 μL of the filtered and enriched supernatant was mixed with 20 μL *S. aureus* RN4220 WT or Δ*tarM* and 4 mL 0,3%-TSA (containing 5 mM $CaCl_2$), after which the agar was poured onto TSA-plates. After incubation overnight at 30°C or 37°C, single plaques were picked and propagated three times to yield pure cultures. This way, several phages were isolated and later confirmed as separate phages via Nanopore sequencing.

Phage DNA isolation

20 mL phage lysate was incubated with 10 μL each of DNAse I (Thermo Scientific, Ref EN0521) and RNAse A (Thermo Scientific, Ref EN0531) for 30 min at 37°C. Next, the lysate was centrifuged in a fixed-angle rotor (Beckman Coulter TI 50.2) at 35.000 x g for 1 h. After discarding the supernatant, the pellet was resuspended in 550 μL TES buffer (0.1 M Tris-HCl pH 8.0, 0.1 M EDTA, 0.3% SDS), and incubated for 15 min at 65°C to disrupt the bacteriophages. Proteins were then precipitated by addition of 60 μL 3 M sodium acetate (pH 4.8), vigorous mixing, and incubation on ice for 15 min. The precipitate was then removed via centrifugation (14,000 x g, 5 min), and the supernatant was mixed with 100 μL phenol:chloroform:isoamylalcohol (25:24:1, pH 7.5). After centrifugation for 5 min at 14,000 x g, the supernatant was combined with an equal amount of isopropanol, centrifuged (14,000 rpm, 15 min), and the resulting DNA pellet was washed twice with 70% ethaol. Residual ethanol was removed by drying at 65°C, and the DNA was resuspended in 500 μL H_2O.

WTA isolation

50 mL of overnight cultures were grown in TSB supplemented with 0.2% glucose, washed in ammonium acetate buffer (10 mM $NH_4CH_3CO_2$, 6 mM $C_2H_3O^-_2$, pH 4.8), and disrupted with glass beads (SM1 special grinding beads 0.1–0.2 mm, Willy A. Bachofen GmbH) in a bead-beating grinder (Precellys evolution touch homogenizer, Bertin technologies) at 5,500 rpm, 5 times, for 30 s (4.5 g glass beads per g of bacterial cell wet weight pellet in several 2-mL screw cap micro tubes, Sarstedt). The acquired cell lysates were incubated at 37°C for 3 h while shaking at 300 rpm in the presence of 5 mM $MgSO_4$, 20 μg/mL DNase (DNase 1 from bovine pancreas, Roche) and 1.6 mg/mL RNase (Ribonuclease A from bovine pancreas, Sigma-Aldrich, R5503). Afterward, sodium dodecyl sulfate (SDS) was added to a final concentration of 2% and incubated for 1 h at 65°C and 350 rpm. Cell walls were washed at least 12 times with ammonium acetate buffer to remove SDS (10 min, 21,000 × g). To release WTA from cell walls, samples were treated with 5% trichloroacetic acid (TCA) for 4 h at 65°C. Peptidoglycan debris was separated via centrifugation (10 min, 15,000 × g). The WTA containing supernatant was adjusted to neutral pH by addition of appropriate amounts of NaOH. WTA was stored at −80°C. Prior to HPLC-MS analysis, 50-μL WTA samples were hydrolyzed with 100 mM NaOH at 60°C for 2 h and subsequently neutralized with appropriate amounts of HCl.

LC-MS WTA analysis

WTA polymer composition was analyzed by liquid chromatography MS (LC-MS) using a micrOTOF-II mass spectrometer (Bruker) coupled to an Ultimate 3000RS high-performance LC HPLC system (Dionex). For HPLC separation (injection volume of 5 μL), a

C18-Gemini column (150 mm × 4.6 mm, 110 Å, 5 μM; Phenomenex) was applied at 37°C with a flow rate of 0.2 mL/min and the following gradient: 5-min equilibration step with 100% buffer A (0.1% formic acid and 0.05% ammonium formate), 30 min a linear gradient of 0–40% buffer B (acetonitrile), for 5 min a washing step with 40% buffer B for and a 5 min re-equilibration step (100% buffer A). Samples were measured in positive ion mode. Exact masses in positive ion mode were presented as extracted ion chromatograms with DataAnalysis (Bruker). Base peak chromatograms were used for sample normalization. LC-MS data processing was done with UmetaFlow GUI (https://github.com/axelwalter/streamlit-metabolomics-statistics) via extracted ion chromatograms with a mass tolerance of 10 ppm.

RBP clustering

A curated RBP list was created by *in-silico* analysis of the genomes of ΦP68, ΦCSA13, Φ11, Φ12, Φ13, ΦK and ΦRemus. These phages were chosen as representatives of their respective cluster due to availability of published data or highest RBP similarity to other phage RBP clusters. The staphylococcal phage database was created by isolation of all phages in the INPHARED database (April 1st, 2023)[31] that contained the keywords "staph", "aureus" or "epidermidis". The curated RBPs were then protein-BLASTed in the staphylococcal phage database, and results were filtered with a cutoff of at least 65% overlap and 65% identity. The results were used for ClustalOmega MSA, which was then processed in SimplePhylogeny and visualized in iTOL.[32,57] The interactive RBP1 tree is accessible under https://itol.embl.de/tree/4652552332411677924980. The interactive RBP2 tree is accessible under https://itol.embl.de/tree/4652552331276168838286 6.

In-silico methods

Clustering was performed with protein blast,[58] ClustalOmega,[32] SimplePhylogeny,[59] and interactive Tree of Life (iTOL).[57] *tarIJL* was discovered by searching for the sequence of the primary *tarIJL* cluster[60] in the bacterial genomes via discontiguos megablast.[58] Analysis of FACS data was done with FlowJo 10.0. Visualization of data and statistical analysis was performed with GraphPad Prims 10.0. PhARIS was developed with Spyder 5.5.0 in Python 3.12.0, the source code can be accessed on GitHub https://github.com/JKrusche1/PhARIS. Validation of PhARIS was performed with the genomes of Rosenblumvirus EBHT, Twortvirus Mallokai, and the newly isolated Triaviruses FK1 and SBK1, as well as Azeredovirinae IS1 and JK1. AlphaFold structures were created with AlphaFold3[33] and visualized with PyMol.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis was done with GraphPad Prism (GraphPad Software, version 10.1.2). Statistically significant differences were calculated by using one-way ANOVA method with Dunnett's multiple comparison or t test. Images show SEM. Value of *n* represents the number of biological repeats. Details of statistical analysis are indicated in figure legends.